



Programming Fundamentals

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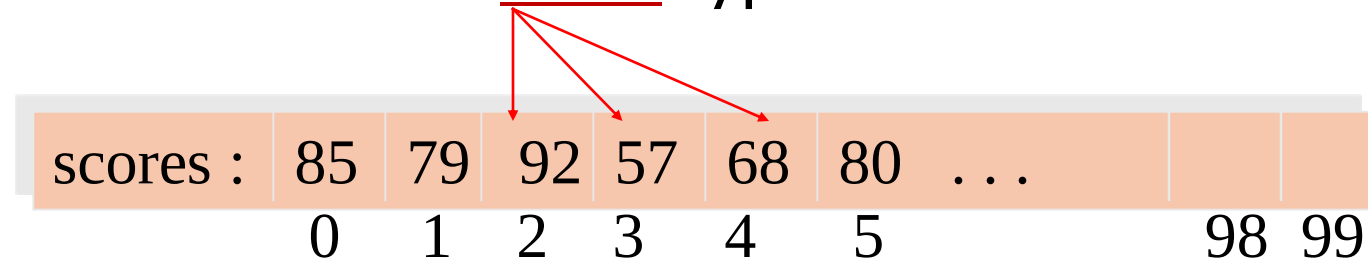
Structures

Structure (Struct)

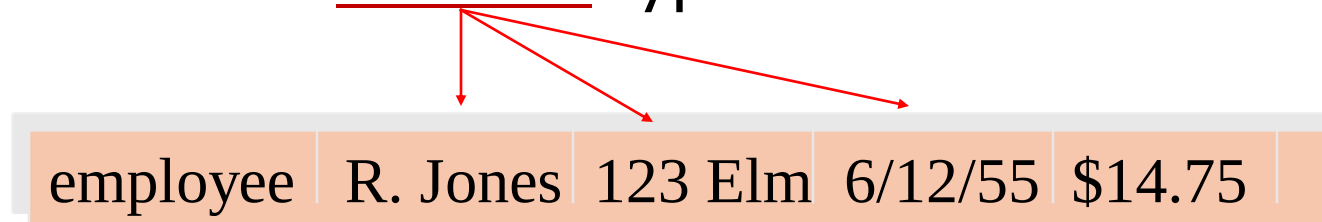
- Arrays require that all elements be of the same data type
- Many times it is necessary to group information of different data types.
 - An example is a materials list for a product.
 - `name` for each item, a `part number`, `dimensions`, `weight`, and `cost`.
- C++ programming supports data structures that can store combinations of `character`, `integer` `floating point` and `enumerated` type data.
- They are called a `structs` (structures)

Structure (Struct)

- Recall that elements of arrays must all be of the same type



- In some situations, we wish to group elements of different types



- Record called a struct (structure)
- Fields called members

Structure (Struct)

- A `struct` is a derived data type composed of members that are each fundamental or derived data types.
- A single `struct` would store the data for one object.
- An array of `struct` would store the data for several objects.
- Structures are referred to as compound data types they consist of several different variables, which are yet logically connected
- A `struct` can be defined in several ways

Structure (Struct)

```
struct my_example
{
    int label;
    char letter;
    char name[20];
} ;
```

/* The name "my_example" is
called a structure tag */

```
struct Person
{
    char name[30];
    int citizenship;
    int age;
};
```

Structure (Struct)

```
struct Student
{
    int StudentID;
    string Name;
    float Batch;
    float GPA;
} Student1;
```

Student1

studentID

Name

Batch

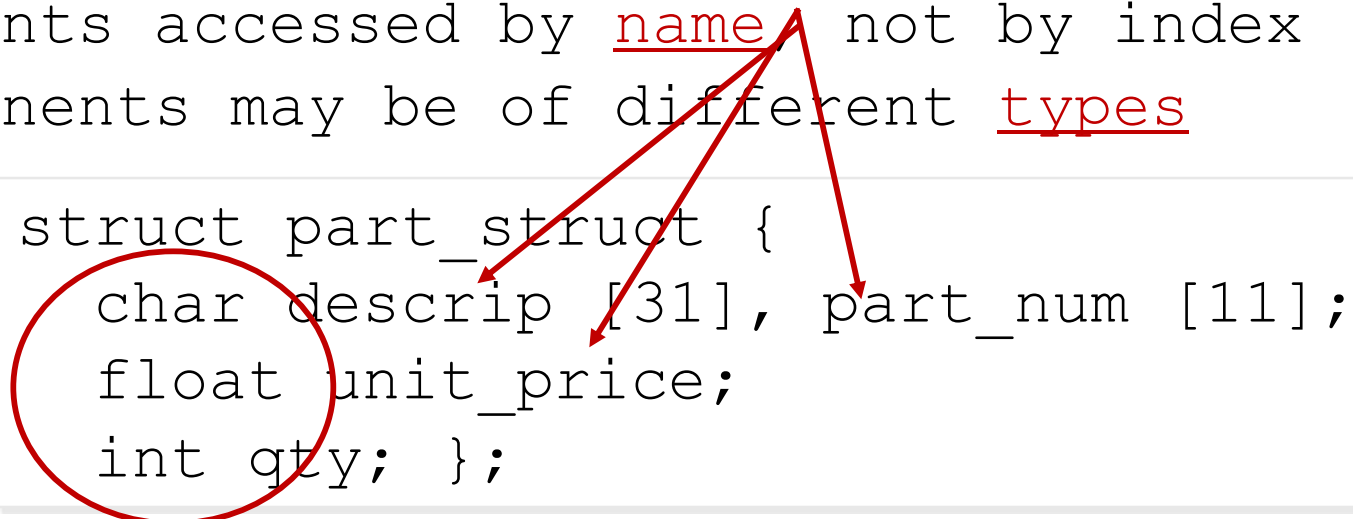
GPA

Structure (Struct)

- C++ **struct**

- structured data type
- fixed number of components
- elements accessed by name not by index
- components may be of different types

```
struct part_struct {  
    char descrip [31], part_num [11];  
    float unit_price;  
    int qty; };
```



Access members of a struct

- Individual members of a `struct` variable may be accessed using the structure member operator (the dot, `"."`)
 - `mystruct.letter ;`
- if a pointer to `struct` has been declared & initialized
 - `Some_name *myptr = &mystruct ;`
 - `(*myptr).letter;`
 - `Employee.name`
 - `Employee.ID`
 - `Employee.salary`

Example of Struct

```
#include <iostream>
using namespace std;
struct Person
{
    char name[50];
    int age;
    float salary;
};

int main() {
    struct Person p1;
    cout << "Enter Full name: ";
    cin.get(p1.name, 50);
    cout << "Enter age: ";
    cin >> p1.age;
    cout << "Enter salary: ";
    cin >> p1.salary;

    cout << "\nDisplaying Information."
    << endl;
    cout << "Name: " << p1.name << endl;
    cout << "Age: " << p1.age << endl;
    cout << "Salary: " << p1.salary;
    return 0; }
```

Enter Full name: Magdalena Dankova
Enter age: 27
Enter salary: 1024.4

Displaying Information.
Name: Magdalena Dankova
Age: 27
Salary: 1024.4

Example of Struct (Initialize)

```
#include <iostream>
using namespace std;
struct Address
{
    string name;
    int number;
    string street;
    string town;
    char state[2];
    string zip;
};
```

```
int main()
{
    Address jd = {"Ahmed", 25, "South St",
                  "New Karachi", {'N', 'K'}, "07974"};
    cout<<"Name: "<<jd.name<<endl;
    cout<<"Street: "<<jd.street<<endl;
    cout<<"State: "<<jd.state<<endl;
    return 0;
}
```

C++ program that demonstrates how to store student details (name, roll number, batch name, and age) in a structure and print them using a function.

```
#include <iostream>
using namespace std;

struct Student {           // Define the
    structure
    char name[50];
    int rollNumber;
    char batch[20];
    int age; };

void printStudent(Student s) { //
    Function to print details
        cout << "Student Details:" << endl;
        cout << "Name: " << s.name << endl;
        cout << "Roll Number: " <<
s.rollNumber << endl;
        cout << "Batch: " << s.batch <<

int main() {
    Student s;
    cout << "Enter Name: "; // Input
    student details
        cin.getline(s.name, 50); // Use
    getline for strings with spaces
        cout << "Enter Roll Number: ";
        cin >> s.rollNumber;
        cin.ignore(); // To ignore the
    newline left by cin
        cout << "Enter Batch Name: ";
        cin.getline(s.batch, 20);
        cout << "Enter Age: ";
        cin >> s.age;

    printStudent(s); // Print the
    details using function
        return 0;
}
```